

Comments on Class Meeting #4

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Some details are important. Please make sure your notes capture the mechanistic details presented in these comments. It is important to learn the correct habits early in the course.

Answers to Selected Problems

Here are some comments on selected problems from today's class meeting.

2. When using isotopically labeled water, the hydrolysis of epoxide gives different products depending on if the reaction is performed in acidic or basic solution. Can you explain the observation by proposing a mechanism for each reaction? How would you exploit this observation in organic synthesis?

The epoxide is asymmetrical. One carbon is secondary and the other is tertiary. In base catalyzed ring opening the hydroxide nucleophile cannot attack at the tertiary centre. Hydroxide attacks at the secondary carbon in a S_N2 substitution reaction. The alkoxide product can then accept a proton from water to regenerate the hydroxide. If the hydroxide that attacked contained an isotope of oxygen, we would see that isotopic label occur at the secondary carbon in the diol product only.

In acid, there are no strong nucleophiles present. The small amount of protonated epoxide can open to relieve ring strain. The tertiary carbocation is much more likely to form than the secondary carbocation, so we will see the vast majority of intermediate be the tertiary cation. Water solvent, which is a weak nucleophile, can add to the strongly electrophilic carbocation. As a result the majority of product would see an isotopically labelled oxygen at the tertiary centre of the diol product.

In a synthesis, we could control where a nucleophilic oxygen added by performing the reaction in acid or in base. Controlling regioselectivity in a reaction is a major goal of synthetic organic chemistry and we will master these principles in this course.

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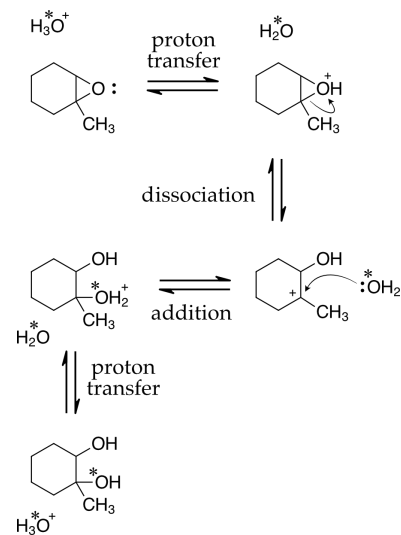


Figure 1: Acid catalyzed epoxide opening

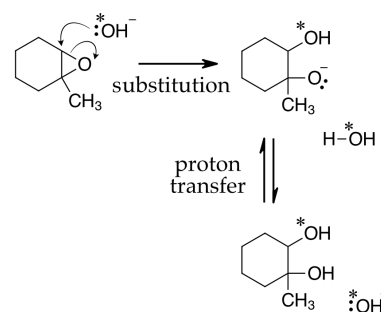
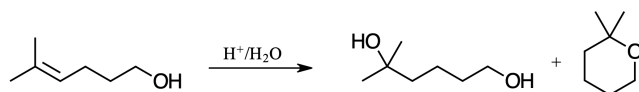


Figure 2: Base catalyzed epoxide opening

3. Explain the mechanisms that give rise to each of the products below. What reaction conditions would you choose to maximize the yield of each?



The acid could protonate the alkene or the alcohol. Protonating the alcohol will lead nowhere because we cannot dissociate the water to create a primary carbocation. Always remember your vow to never propose an unstabilized primary carbocation. Protonating the alkene would give a tertiary carbocation that can have water add to give the alcohol product or have intramolecular ring closing to give the cyclic ether.

To maximize the yield of the diol we should perform the reaction in water. To maximize the yield of the cyclic ether we should perform the reaction in organic solvent using a non-aqueous acid such as sulphuric acid or TsOH

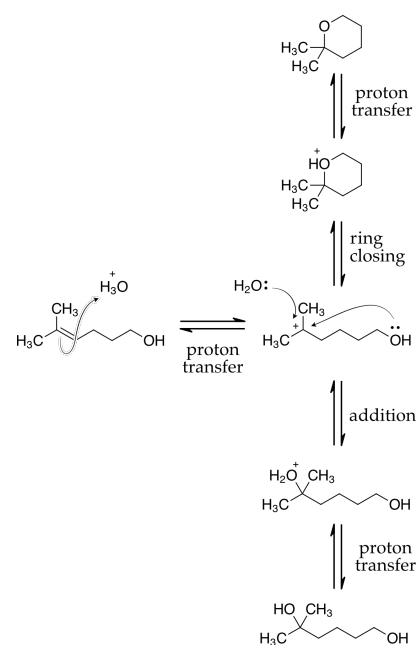


Figure 3: Acid catalyzed addition to an alkene

4. Bromination of acetone in water is fast but the rate is independent of bromine concentration. (See figure 4). Can you explain the observation by proposing a mechanism for the reaction?

Bromine reacts rapidly with alkenes. The reactive species in this reaction is the enol tautomer of acetone. Enols are electron-rich and will react rapidly with bromine. It is the formation of the enol that is the rate-determining step followed by the rapid bromination reaction. So the rate law is zero order in bromine concentration.

The first step is the tautomerization to form an enol. The equilibrium constant for the enol/ketone equilibrium of acetone is 8×10^{-8} . Only a small amount of the enol will be present, but it is an electron-rich alkene and reacts with the bromine quickly. The rate of formation of the enol is often the rds.

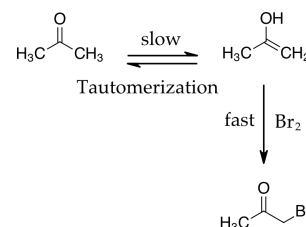


Figure 4: α -bromination of acetone

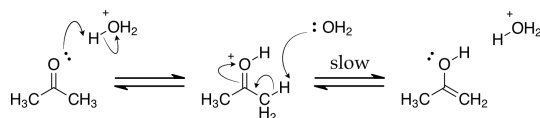


Figure 5: Tautomerization of a ketone to form an enol

The concentration of bromine does not affect the rate in water. Take note of the fact that the rate of α -halogenation is the same whether chlorine, bromine or iodine is used as the reactant.

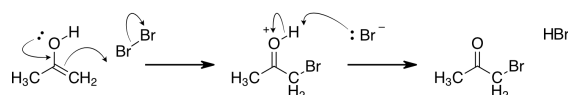
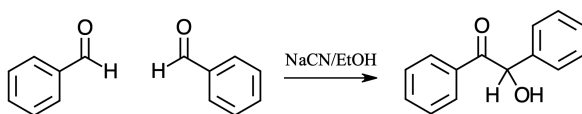


Figure 6: Electrophilic bromination of the electron-rich double bond of the enol.

5. The benzoin condensation seems to be impossible. And it is unless catalyzed by cyanide ion. Propose a mechanism. Be able to argue for the stability of your intermediates.



You may think that the benzoin condensation can be explained by the cyanide anion acting as a base to deprotonate the carbonyl group of the benzaldehyde and make a nucleophile as shown in figure 8. The anion that we imagine this way is a very high energy anion. The lone pair is not stabilized and this species is not possible (at least it has never been observed.) Yet we observe reactions like this in biochemistry all the time.

The mechanism is shown in figure 9. The cyanide ion adds to the carbonyl group of the aldehyde. The adduct can be deprotonated because the resulting negative charge can be delocalized through the π system of the cyano group (figure 10.) We do not have a deprotonated aldehyde, we have a deprotonated cyanohydrin. This is the nucleophile that can add to another benzaldehyde to give an aldol product. The cyano group is finally eliminated to regenerate the carbonyl group and the cyanide anion. The cyanide is a catalyst.

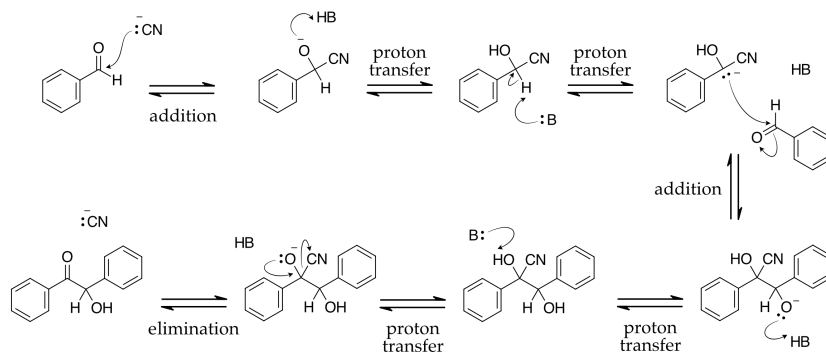


Figure 7: The benzoin condensation

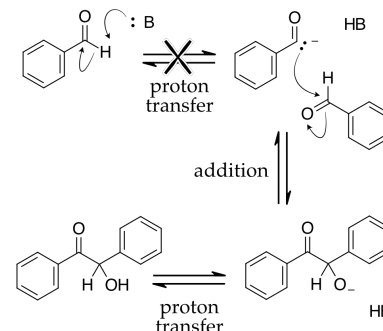


Figure 8: An impossible mechanism

Figure 9: The mechanism of the cyanide-catalyzed benzoin condensation

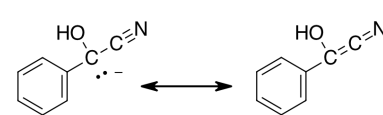


Figure 10: Stabilization of the cyanohydrin anion